

Building a Scalable Compliance Data Warehouse for a Global Asset Management Firm

Introduction

A U.S.-based global asset management firm managing approximately \$500 billion in assets under management (AUM) partnered with NuWare to develop a unified, enterprise-grade data warehouse solution to streamline compliance surveillance, regulatory reporting, and internal audit operations.

The client operates multiple trading systems across global markets, handling both cash and derivatives transactions. These systems generated massive volumes of data, stored across siloed platforms; making it increasingly difficult for the firm's compliance and audit teams to retrieve, validate, and report trading information in a timely and consistent manner.

NuWare's Argus Data Warehouse Solution was designed to consolidate disparate trade data sources, enhance reporting accuracy, and ensure global compliance adherence, while maintaining scalability for high-volume financial operations.

The Challenge

The firm's compliance and surveillance division faced several complex challenges in managing transaction data and meeting regulatory expectations:

- 1. Fragmented Data Landscape**

Transaction data resided in multiple systems, separate ones for cash, stock, program, and options trades; with each market (U.S., U.K., Asia) maintaining independent repositories.

- 2. Regulatory Reporting Complexity**

Different regulatory bodies required distinct reporting formats, resulting in heavy manual intervention and duplicated efforts.

- 3. Massive Data Volume**

Over 250 million transaction records were received daily for order execution comparisons against market quotes; a major challenge for analysis and retention.

- 4. Inconsistent Order Flow Tracking**

The firm lacked an integrated mechanism to trace order hierarchy or transaction lineage across trading systems.

5. Audit and Compliance Bottlenecks

Responding to requests from regulators or internal audit teams required extensive cross-system searches, delaying compliance timelines.

These inefficiencies not only increased operational costs but also created potential risks of delayed or inaccurate compliance reporting, impacting the firm's reputation and regulatory standing.

NuWare's Approach

To address these challenges, NuWare engineered a high-performance Data Warehouse and Reporting Platform, internally known as AB Argus; designed to handle large-scale, cross-market data ingestion, validation, and analytics.

1. Data Integration and Consolidation

- Developed an enterprise-grade data warehouse capable of integrating and loading data from multiple Order Management Systems (OMS) used across trading desks.
- Incorporated market quote feeds from vendors and regulatory reports into a single unified repository.
- Established data lineage tracking by capturing entity hierarchy and order flow structures across systems.

2. Performance and Scalability

- Architected a data processing system optimized for T+1 data availability to compliance teams.
- Implemented compressed data storage techniques, minimizing infrastructure costs while supporting up to 10 years of historical data retention.
- Tuned for high-speed query execution, ensuring real-time analytics even under multi-terabyte data volumes.

3. On-Demand Reporting and Advanced Querying

- Delivered a custom application interface for compliance users to generate on-demand reports and conduct ad-hoc queries using an "Advanced Query" module.
- Enabled comparison of order executions with Best Bid/Ask prices at the time of trade.

- Provided multi-level search capabilities for trade and order flow tracing within seconds.

4. Compliance and Audit Enablement

- Integrated workflows to support regulatory reporting and internal audit validation.
- Automated tracking and comparison of data submissions made to various regulatory agencies.
- Supported multi-region compliance frameworks, accommodating local data privacy and storage mandates.\

Outcomes

The AB Argus solution revolutionized the client's compliance monitoring and data management processes, delivering measurable business and operational benefits.

1. Unified Compliance Data Access

- Consolidated data from multiple OMS and trading systems into one centralized repository.
- Enabled a single source of truth for compliance, audit, and reporting teams.

2. Enhanced Efficiency and Accuracy

- Reduced manual report preparation effort by over 50%.
- Automated compliance workflows minimized human error and ensured audit readiness.

3. Faster Regulatory Response

- Improved turnaround time for regulatory inquiries from weeks to hours.
- Provided real-time search and analytics across millions of records.

4. Improved Governance and Data Transparency

- Established clear data lineage and entity hierarchy for every order flow.
- Empowered internal audit teams to perform self-service queries and validations.

5. Cost and Performance Optimization

- Achieved compressed data storage with up to 40% cost savings in infrastructure maintenance.

- Maintained T+1 data availability while handling multi-terabyte daily inflows without performance degradation.

Key Features

- **T+1 Data Availability** for compliance teams.
- **10 Years of Historical Data** stored securely and efficiently.
- **Advanced Query Engine** for ad-hoc, multi-dimensional searches.
- **Multi-Region Scalability** across trading desks (U.S., U.K., Asia).
- **Compressed Data Storage** to minimize costs without compromising speed.